Physics Olympiad Errata

This file lists all the errors in Physics Olympiad solutions that I am aware of. Even at the international level, solutions files have more errors than one would think. People can spend a lot of time searching for a mistake in their work, when in fact it's the official solution that is wrong. My hope is that having all the errors in one place will make it easier for students to go through the problems by themselves.

Some items in the list are marked with a blue dot (•). These should be read before attempting the respective problem – they either serve to clarify ambiguities in the problem statements, or they warn about completely broken problems. The remaining items mostly cover errors in the original solutions, and should be treated as spoilers. All problems and solutions can be found on phoXiv.

IPhO

For reference, I have checked all the papers up to 1984. I do not recommend the problems from before that, as their style differs too much from modern Olympiads. They're more open-ended and they cover subjects that are less common nowadays (for example, lenses and kinetic theory).

(1) 1967 3 The original solution implicitly assumes that the two balls do not deform at all due to gravity. This turns out to be inconsistent, because then you would be able to leverage thermal expansion to violate the second law of thermodynamics. The correct solution is a bit more subtle but still accessible to a high-school student. See here.

(2) 1971 1B Equation (13) should be corrected to

$$-m_2 a \sin\alpha_2 = T \sin\alpha_2 + R_2 \cos\alpha_2 - m_2 g.$$

Everything that follows is alright, though they've missed a g in the answer for a_0.

(3) 1972 3 Here you need to find the maximum height that the liquid will reach during the subsequent motion, not the height at which it will settle after a long time due to frictional losses. Note also that the capacitor is isolated. •

(4) 1983 2C An error in the calculations. Defining $C \equiv C_1 + C_2$ and $L \equiv L_1 L_2/(L_1 + L_2)$, you should obtain

$$I_{A\to B} = \frac{L_1 I_{01} - L_2 I_{02}}{L_1 + L_2} + C_1\sqrt{L^2(I_{01}+I_{02})^2 + (LC)u_0^2} \times \left(\frac{1}{LC} - \frac{1}{L_1 C_1}\right)\cos\left(\frac{1}{\sqrt{LC}}t - \arctan\left(\frac{I_{01}+I_{02}}{C' u_0}\sqrt{LC}\right)\right).$$

Since $L_1 C_1 = L_2 C_2 = LC$, only the constant term will survive, and we have $I_{A\to B} = -0.1\,\mathrm{A}$. If you manage to spot from the start that the numbers are nice, you can quickly solve the problem heuristically (but note that you'll only get the numerical answers, not the general formulae). See here. •

(5) 1985 1B To clarify, both the angles are measured clockwise from the north. •

(6) 1986 1.1 A typo in the problem statement – they mean $A = 2a\cos\beta$ rather than $A = 2a\cos\theta$. •

(7) 1986 1.5 The solution for this subpart doesn't match the question. I suppose that the original question was to find the values of β for the zeroes and the adjacent-to-principal maxima, as well as to determine the intensities of the adjacent maxima.

(8) 1986 2.1 The inequality symbol should be reversed, $\theta < \arccos(R_C/R)$. •

(9) 1986 2.3 A typo in the data, $v_P = 10.85\,\mathrm{km/s}$. •

(10) 1986 2.3 On the graph for the travel time of the rays as a function of θ, the refracted rays should get multiple branches – one for each incidence angle i that may result in a given θ. For the correct graph, see this page.

(11) 1986 3.3 Note that all the amplitudes a_s are equal. •

(12) 1986 3.4 The final results in the original solution are correct, but their intermediate calculations are incoherent. Start instead with

$$\frac{u_n}{u_{n+1}} = \frac{\sin\left(\frac{2ns\pi}{N} + \varphi\right)}{\sin\left(\frac{2(n+1)s\pi}{N} + \varphi\right)}.$$

In case (a) the counter s is small, so the denominator leads the numerator by a tiny phase shift $2\pi s/N$, but otherwise $(u_n/u_{n+1}) \approx -1$.
In case (b), $s = N/2$ when N is even and $s = (N-1)/2$ when N is odd. Then, for even N the denominator leads by exactly π, so $(u_n/u_{n+1}) = -1$. For odd N, the ratio is almost -1, but there's also a tiny phase shift $-2\pi/N$. This shift is less than the one in case (a), so as you can see on the graphs in the original solution, you need to count out many more particles until you accrue a total phase shift of 2π.

(13) 1988 2 There are many typos and calculation errors here, but even worse, the original solution for the fourth subpart is entirely wrong. It's not possible for the wheel to move horizontally in the absence of horizontal forces. The geometry of the problem is rather more complicated and it gives rise to some interesting physics. Use my rewrite here. •

(14) 1988 3.2 The energy equation in the solution should also have an $h\nu$ on the right hand side.

(15) 1988 3.3 Be careful – in the solutions for this subpart and the next, there are lots of typos in the denominators of the energy terms.

(16) 1988 3.5 Change the photon's angular frequency from $2.057 \times 10^{17}\,\text{rad/s}$ to $2.607 \times 10^{17}\,\text{rad/s}$, as that's what the original solution works with.

(17) 1991 2A Equation (4) in the solution should read $a_{\text{AB}} = \frac{\sqrt{1-(v/c)^2}}{1+(uv/c^2)}a$.

(18) 1991 3A Part of the problem statement is missing – the complete phrase is "by those atoms whose kinetic energy is equal to the mean kinetic energy of the atoms inside the region behind the collimator". •

(19) 1991 3C Here you're asked to find the maximum value of φ. You may use approximations. •

(20) 1991 3C A calculation error, $\varphi = 2.8 \times 10^{-5}\,\text{rad}$.

(21) 1991 3E To solve this part, you need to also make use of the fact that the probability distribution for the re-emission angle θ is isotropic (i.e. the direction of re-emission is random). Certain other problems on Doppler laser cooling instead work with an oversimplified model where re-emission can only occur along the axis of the laser beam (IPhO 2009 2, IPhO 2024 2B). My preferred problem on the subject is APhO 2006 1, which doesn't cut corners like that. •

(22) 1991 3E One more calculation error, $\Delta t = 3.4 \times 10^{-3}\,\text{s}$.

(23) 1992 1 This is an instructive problem on tidal forces, but its wording is a bit too vague. I suggest that you use the following improved versions of the problem statement and the solution, edited by Teo Kai Wen. You'll still need the original solution for reference though. I've uploaded it here. •

(24) 1993 1.4 This is a real-world problem for which I don't think the assumptions are made clear enough. Firstly, you'll need to assume that the field right above the quadrants always remains equal to E_0 (say, due to continuity, as derived from $\oint \mathbf{E} \cdot \mathrm{d}\mathbf{l} = 0$). This should come above all other considerations. •

(25) 1993 1.5 The crux of the problem is what happens to the charge on q_l the lower surface of the insulated quadrants – after all, the total charge on the quadrants that accumulates through the capacitor and the resistor is actually $q + q_l$. To resolve this issue more easily, you can assume that the ground is quite close to the quadrants, at a distance $l \ll r_2$. See below for more details.

Consider a line integral of the electric field $\mathbf{E}$ in the air, starting from the lower surface and ending at the ground. This gives us $V = \int_{\text{lower}}^{\text{ground}} \mathbf{E} \cdot \mathrm{d}\mathbf{l}$. As you can see from the diagram, the ground is in close proximity (it's the ring around the rod), so from Gauss's law $E \approx q_l/(\varepsilon_0 S_{\text{quadr}})$ and $V \approx El$, where l is some characteristic distance. Notice that $\varepsilon_0 S_{\text{quadr}}/l$ can be used as an order-of-magnitude estimate for the capacitance of the quadrant system C_s when you take out R and C and all that. Therefore $V = q_l/C_\text{s}$, and since we're told that C_s is small, we can conclude that $q_l \approx 0$. ?

(26) 1994 3.1 The minus sign on the right hand side of Equation (5) in the solution should be a plus. No errors after that.

(27) 1995 1C.2 They've made an error when converting the units for the rest mass. The numerical answer ought to be $\Delta f/f_0 = 5.44 \times 10^{-9}$.

(28) 1998 3F Referring to the notation of Figure 3.2, the correct formula to use for the Doppler shift is $\lambda = \lambda_0\gamma(1 + \beta\cos\phi)$. But this doesn't change the final answer.

(29) 2000 3B I think this warrants a clarification; here ω is the lowest permitted angular frequency of a longitudinal standing wave in the rod, given the boundary conditions. •

(30) 2000 3D For this part and the next, assume that Δg is directed along one of the rods. •

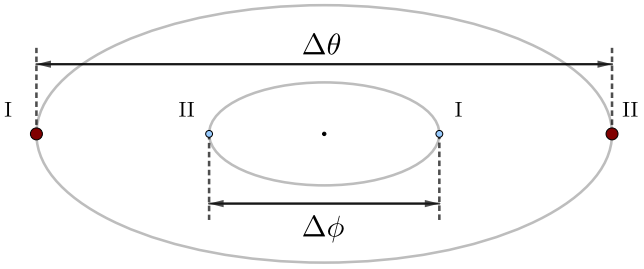
(31) 2000 3E I believe they mean that the minimum detectable optical path difference is 10^{-4} of the laser's wavelength. In that case, a displacement Δl in one rod brings about an OPD of $2\Delta l$ because the light goes back and forth. The answer should therefore be twice as large. •

(32) 2000 3F There's no official solution. My answer is $\sqrt{300/0.1} = 55$, which I got through some handwaving with energy scales, as in $kA^2 \sim mv^2 \sim k_B T$. ?

(33) 2000 3G No official solution again. The answer should be, hopefully in clear enough notation,

$$\tau = \frac{ms_i T_i}{4P}\left(1 - \left(\frac{T_f}{T_i}\right)^4\right) = 25\,\text{h}.$$

(34) 2001 2A You'll also need to assume that the orbits are circular and tilted with respect to the line of sight. A better diagram for how the orbits are projected on the night sky would look like this: •



(35) 2001 2A To put it another way, they're telling you that the time interval between betweeen configurations I and II is equal to τ.

(36) 2001 2B Both the problem statement and the solution are incoherent. Here's a rewrite...

> In what follows, assume that $M \gg m_0$ and denote the radius of the ordinary star's orbit with respect to the neutron star by r_0. The ordinary star ejects a blob of gas. In the reference frame of the ordinary star, the velocity of the gas outflow is directed towards the neutron star and is equal to v_0. Find the distance of closest approach r_f between the gas blob and the neutron star. Neglect the gravitational influence of the ordinary star.

The answer that I get is $r_f = \frac{r_0}{1+v_0\sqrt{r_0/GM}}$.

(37) 2001 3 The solutions file includes some tasks which were removed in the final version of the problem statement. You may disregard those. For an extension of the problem with a cleaner solution, see the write-up here.

(38) 2002 3 The original solution is so badly formatted that it is rendered illegible. There are also major conceptual and algebraic errors. You can instead use the following edited version of the problem.

(39) 2003 1C The answer for α_c in the marking scheme should be to the power of -1, just like in the solution.

(40) 2004 3A There is a minor typo in the expression for $\tan\phi$ in the answers section. The correct answer is $\tan\phi = \frac{b\omega}{m(\omega_0^2-\omega^2)}$, as derived in the detailed solutions.

(41) 2005 2.7 Read Footnote 2 as "When you're calculating the force on the coils, you can neglect the misalignment between their axes."

(42) 2008 1.3.1 Note that in this subpart you are allowed to approximate. In particular, when you're computing the moment of inertia of the water, you should consider it as a point mass.

(43) 2009 2.3D The expression for ε_{at} is incorrect, you should instead find that $\varepsilon_{\text{at}} = \frac{mv^2}{2}\left(1 - 4\frac{\hbar q}{mv}\right)$. This error propagates throughout the rest of the problem. Here are the rest of the answers that need corrections. In **4C**, $\varepsilon_{\text{at}}^{-} = \frac{mv^2}{2}\left(1 - 2\frac{\hbar q}{mv}\right)$. In **5A**, $\Delta\varepsilon = -\hbar\omega_L\frac{v}{c}$. In **6A**, $\Delta\varepsilon = +\hbar\omega_L'\frac{v}{c}$.

(44) 2011 1.3 This part of the problem is wrong because the equilibrium about the Lagrange point in question is unstable. You can see this by considering the force on a stationary object displaced radially from the Lagrange point by a little bit (in the reference frame of the rotating objects). You will find that this force points away from the Lagrange point. The additional assumption of constant angular momentum is both false and redundant. See this article by Jaan Kalda for a detailed analysis.

(45) 2014 2B.8 The value for the surface tension is $\sigma = 0.12\,\text{N/m}$ (without the 10^{-2}).

The questions on the 2017 paper are very unclear. My comments below refer to the original problem statements, but instead of working with those, I suggest that you just use the corrected version here, made by Sirawish Musikasopon.

(46) 2017 2B.1 Depending on how you measure the arrival time, your answers in the next two parts may diverge significantly from those in the original solutions. So, to be explicit – assume the arrival of the P-wave corresponds to the instant when the seismometer reading becomes nonzero.

(47) 2017 2B.2 Another comment – do not take any additional measurements for this part. Use only your results from **B.1**.

(48) 2017 2B.3 And yet another clarification – assume that the first signal to arrive at DNP is indeed due to the wave travelling through the mantle.

(49) 2017 2B.4 The original solution is incomplete. We need to take the upper limit of the integral for X, and only then do we find $X = \frac{2}{ap}\sqrt{1-(pv_0)^2}$.

(50) 2017 2B.6 The result of this calculation is very sensitive to the denominators, and the original solution hasn't been careful with this. I get $T = 184.1\,\mathrm{s}$.

(51) 2017 3A.5 In this subpart, R_0 is not the current value of $R(t)$ as the problem statement after **A.2** would suggest. There's something else that happens first – assuming that the arbitrary constant k in Equation (1) is not zero, we take the curvature term $-kc^2/R(t)^2$ and we rescale the numerator and denominator by the same positive value such that k is now normalised to either 1 or -1. Now, the denominator is some other radius $R'(t)$ squared. We call the present value of that radius R_0 (and therefore $R'(t) = R_0 a(t)$).

(52) 2017 3B.1/3B.2 To clarify, here you're supposed to plug in the $k = 0$ no-curvature solution from **A.3** to study the case of small curvature (i.e. $k = \pm 1$ and large R_0). This is not strictly correct, but it's still informative as it enables us to get a sense of the time evolution of $\Omega(t) - 1$. You're supposed to find that it can be made to tend to zero so long as energy density dominates. Hence the need for inflation in the early universe.

(53) 2017 3B.3 The "condition for inflation" you are expected to use is $w = -1$, not anything that comes later in the text.

(54) 2017 3D.2 To be a bit more rigorous than the official solution, the observational constraints are $-5.19 > n > -6.69$ and $n > -1.81$, and these cannot be satisfied simultaneously.

(55) 2022 2C The model used here isn't self-consistent (as discussed in the solutions), and I think it's impossible to figure out what you are supposed to do if you are attempting the problem on your own. The rest of the problem is really nice.

(56) 2026 1C.1 The final answer for $\omega_{\min}$ in Solution 2 is missing a factor of two in front of ΔU.

EuPhO

(57) 2019 2 The diagram in the solution is misleading. The ball can roll without slipping even if the vertical component of the angular velocity of the ball $\boldsymbol{\omega}$ is nonzero. It's just that the contact points between the ball and the table will then trace out an arbitrary tilted circle on the ball rather than a vertical great circle. In this general case, the vertical component ω_z remains constant because there are no torques along z, while the horizontal component always has to point perpendicular to $\mathbf{v}$, and behaves as described in the solution. The parameters of the precession do not change.

(58) 2024 3B There are a few typos in the intermediate calculations in Solution 1. It should read:

$$(|r|^2 + |t|^2)^2 = |r|^4 + |t|^4 + 2|r|^2|t|^2 = 1$$

$$\implies -2|r|^2|t|^2 = r^2 t^{*2} + r^{*2} t^2.$$

And "This is equivalent to $(rt^* + r^* t)^2 = 0$". The rest is fine.

(59) 2024 3B Solution 4 is alright conceptually, but the energy fluxes I_{in} and I_{out} have not been computed correctly – after all, we want to obtain $\sin(\pi + 2(\theta - \psi)) = 0$ instead of $\cos(\pi + 2(\theta - \psi)) = 0$. Unfortunately, the calculations turn out to be quite tedious.

(60) 2026 3 Contrary to what the problem statement says about all surfaces being smooth, you need to assume that the metal disk will stop moving immediately should it touch the ground.

APhO

I have double-checked everything up to 2009.

(61) 2001 2.1 We aren't told anything about the plane of rotation in this part of the problem, so for now you should stay agnostic about the expression for the angular momentum about the CM – just write $\mathbf{L}_{\text{rot}}$ instead of $I\omega$.

(62) 2002 1 There's a typesetting error in the solution. Any time you see a "_" there, keep in mind that it stands for $\frac{1}{2}$.

(63) 2003 2A The solution is incorrect because it doesn't account for the relativistic correction on the speed of light in a moving medium, which is of order $\Omega R/c$ too. This is the same correction as the one in Fizeau's experiment (for that, see Wikipedia or IZhO 2018 3). The final result should be $\Delta t = \frac{4\pi R^2 \Omega}{c^2}$ as per this paper – the refractive index doesn't matter.

(64) 2003 2B $\Delta L = c'\Delta t$ is not the correct expression for the optical path difference. The OPD is what you multiply by the wavevector in vacuum k to get the phase difference $\Delta\phi$, so

$$\Delta\phi = \omega\Delta t = k\Delta L \qquad \Rightarrow \qquad \Delta L = \frac{\omega}{k}\Delta t = c\Delta t.$$

Then in the next part we have $\Delta L = 3.0 \times 10^{-12}\,\text{m}$. After that we obtain

$$\Delta\theta = N\omega\Delta t = \frac{8\pi^2 R^2 N\Omega}{c\lambda}.$$

(65) 2003 3E You need to assume that the plasma slab is a cylinder coaxial with the electron beam. You'll also have to take it on trust that after a long time, all the electrons in the plasma get blown out, while the ions stay in place (this isn't immediately obvious, but it's an experimental fact). Note that most of the solution is redundant, you only care about the expression for the force on an ion.

(66) 2010 1Ab.i I think that the phrase 'only vibrational and rotational motions can be excited' should be disregarded because it hints at an elastic collision, which is not what happens in the original solution. In fact, I don't think that an elastic collision can even occur within this toy model, because the immediate response of the pivoted spring is inevitably associated with some heat losses (imagine the same situation if there was a rigid rod instead of a spring).

(67) 2010 1Ab.i In order for your solution to match that of the problem author, you need to think of Q as the energy stored in the oscillator after the collision. Note that this Q doesn't have that much in common with the Q of part (a).

(68) 2010 3Ad Note that in this subpart we're still working with $E_{\text{k}} = E_0$.

(69) 2011 2G Make sure that you're working with the correct version of the solutions file. Some problem repositories offer an older version which has mistakes. For reference, the value of Ω in the correct file is $5.6 \times 10^{-3}\,\text{s}^{-1}$.

(70) 2014 2D.ii This is like IPhO 1993 3, but much less clear. Here you must assume that the electrons pass quite close to the wire, such that the angle between the rays coming from S is negligible compared to the deflection angle due to the wire. In other words, simply assume that the rays from S are parallel before approaching the wire.

(71) 2014 2D.v To clarify, they're asking you to find the distance between two imaginary slits placed next to S which would give rise to an identical interference pattern.

(72) 2015 2C The ratio of the magnetic fields is 0.3, not 1.0.

(73) 2015 2D.ii The inequality should go the other way, $\theta_{\text{cr}} \leq \arcsin\left(\sqrt{\frac{B_0}{B_m}}\right)$.

(74) 2018 3A.2 For this part only, you should also assume negligible thermal conductivity.

(75) 2019 2A.8 They haven't evaluated the integral correctly. I get $F_{\text{Pr}} = 7.55 \times 10^{26}\,\text{N}$. Then, the answer to A.9 is 152 %.

(76) 2019 2C.1 The $\sin\phi$ factor in the cross product is redundant, and thus $\Omega = \frac{eB}{\gamma m}$.

(77) 2019 2C.2 The beam makes an angle ϕ to the rotation axis, so it doesn't just sweep a great circle on the celestial sphere. The beam will subtend an angle of $\frac{2}{\gamma \sin\phi}$ as seen from the rotation axis (not the origin!), and so $\Delta t = \frac{2}{\gamma\Omega\sin\phi}$. The rest of the solution is fine.

(78) 2019 2D.2 The original solution is flawed because the power law distribution isn't actually unbounded (i.e. from 0 to ∞). Instead, we should consider the following. The particles that end up in $[\varepsilon,\, \varepsilon + \mathrm{d}\varepsilon]$ after the expansion originate from the range $\left[\varepsilon\left(\frac{V}{V_0}\right)^{1/3},\, (\varepsilon+\mathrm{d}\varepsilon)\left(\frac{V}{V_0}\right)^{1/3}\right]$. The number of particles there is

$$V_0 f_0 \left(\varepsilon\left(\frac{V}{V_0}\right)^{1/3}\right) \mathrm{d}\varepsilon \left(\frac{V}{V_0}\right)^{1/3}.$$

We can obtain $f(\varepsilon)\mathrm{d}\varepsilon$ through dividing this by V. The result is

$$f(\varepsilon) = \left(\frac{V}{V_0}\right)^{-\frac{2+p}{3}} \kappa_0 \varepsilon^{-p}.$$

(79) 2021 1C.2 For your working to match theirs, from this point onwards you should neglect β and ignore the terms in α that aren't just a number.

(80) 2021 1D.2 Some of the numerical answers are a bit off, you should get $\rho c v_1 = 5.3\,\text{MPa}$ and $\rho c v_2 = 3.7\,\text{MPa}$.

(81) 2021 2B.3 There's a typo in the third line of the solution. The components of E are $E_z = E\cos\gamma$ and $E_\perp = E\sin\gamma$, where γ is the angle that $\mathbf{E}$ makes with the z-axis (in **B.2**, angle γ would've been equal to $90° - \theta_1$). Otherwise the solution is fine.

(82) 2023 1 To allow for easier comparison with the original solution, I advise you to work in terms of g_0, R_E, and z_h from **B.1** onwards. In particular, avoid expressions that include G, M_E, and the altitude h.

(83) 2023 1A.4 To clarify, you should assume that the temperature is $T = 425\,\text{K}$ everywhere, including at ground level.

(84) 2023 1C.2 Here the task is to simplify the integral from **B.5** to an analytically tractable form and then evaluate it.

(85) 2023 1E To solve this part of the problem, some additional information is needed first. Assume that the oxygen is singly ionised. Assume also that the Earth has a dipole magnetic field such that the magnetic poles coincide with the geographic poles. The magnitude of the magnetic field near the ISS should always be taken for $B = 5 \times 10^{-5}\,\text{T}$. The angle between the local magnetic field and the velocity of the ISS is denoted by ϕ.

(86) 2023 1E.1 I find the problem statement too vague, and I think some more guidance is warranted. Here's how I understand it – the idea is that both the front and the back of the station have reached some equilibrium charge density. The front attracts electrons with an effective capture cross-section S, while the back attracts ions, again with a cross-section S. The internal current serves to maintain the equilibrium by compensating for the loss of charge density due to recombination. ?

(87) 2023 1E.2 As a warning, you will find that the braking force F_{ind} is perpendicular to the velocity of the ISS rather than antiparallel. You are supposed to assume that this quantity is still a good order-of-magnitude estimate for the antiparallel braking force in the real-world system, which is wildly complicated. ?

(88) 2023 1E.2 Bear in mind that Equation (31) in the solution is just an estimate. For an exact expression, you'd need to average out ϕ with respect to time, not with respect to angle.

(89) 2023 1E.3 Equation (35) in the solution needs an extra n_{ion}.

(90) 2023 1F I've made my own answer sheet where I corrected the erroneous numerical answers in the original solution.

(91) 2023 2C.2 Your answer for the acceleration $\mathbf{a}$ should be in terms of m, r, I, Q, B, Ω, the position $\mathbf{R}$, and the velocity $\mathbf{v}$. •

(92) 2023 2C.3 The β' that they refer to is defined as the coefficient in front of ρ in the ordinary differential equation for $\rho(t)$, scaled such that the coefficient of the highest derivative is unity. •

(93) 2023 2C.3 An extra Ω is needed in the numerator of Equation (86) in the solution.

(94) 2023 2C.3 You will find that the motion should simultaneously satisfy

$$\ddot{\rho} + (\beta - \gamma\eta - \dot{\eta}^2)\rho = 0, \qquad \dot{\rho}(\gamma + 2\dot{\eta}) = 0, \qquad \rho\ddot{\eta} = 0.$$

Apart from the solution for $\rho(t)$ and $\eta(t)$ in the official files (which can be made to fit any no-slip initial conditions), there is a trivial solution $\rho = 0$, as well as a special case where ρ is constant and $\dot{\eta} = \frac{1}{2}(-\gamma \pm \sqrt{\gamma^2 + 4\beta})$. For $\Omega = 0$, this indeed matches the answer to EuPhO 2019 2 – for a charged spherical shell where $I = \frac{2}{3}mr^2$, both setups will yield $\dot{\eta} = 4QB/5M$.

(95) 2023 2C.4 There's a missing piece of information without which you cannot proceed. In this subpart, we assume that the parameters of the system are chosen so that $\beta' = 0$. •

(96) 2023 2C.4 The charge of the ball will turn out to be positive.

(97) 2023 2C.5 Work with $I = 0.1\,\text{kg}\,\text{m}^2$ as given by the formula in the problem statement. If you choose to work with their value $I = (1/11)\,\text{kg}\,\text{m}^2$, you'd need to recalculate Ω, which they don't do in the official solutions. •

(98) 2023 2C.5 The final answers are wrong due to a sign error when taking the scalar product. We should instead get

$$\Delta E = 2\pi^2 N^2 \left(\frac{(1+\Omega)^2}{10} + 1\right), \qquad \Delta E_N = 4\pi^2 N \left(\frac{(1+\Omega)^2}{10} + 1\right),$$

where Ω can be either of the two negative values found in the previous subpart.

(99) 2023 3 Confirm that you're working with the most recent version of the official solution – older versions have all the numerical answers wrong. You're using the right file if the pressure in Equation (4) is evaluated at 113 kPa.

(100) **2023 3B.3** Because the jump in the pressure is large, you may work under the assumption that the minimum radius of the bubble is much smaller than its initial radius. •

(101) **2023 3B.3** You should end up with $R = 2.9\,\mu\text{m}$.

(102) **2023 3C.1** The goal here is to obtain two separate expressions for the mass loss from the bubble $\mathrm{d}m/\mathrm{d}t$, one in terms of R' via the ideal gas equation, and one in terms of $\partial u/\partial r$ via Fick's law. We want to equate these expressions to get an ordinary differential equation for R, but to simplify the partial derivative above, we first need to determine $u(r,t)$. This is extremely difficult.

In the official solution, the problem author derives the relevant partial differential equation for $u(r,t)$ and then solves it with a few hard-to-spot changes of variables. But while doing this, he takes the liberty to simplify the maths with an unphysical assumption (temporarily taking the radius R as constant). So, I think that his result for $\partial u/\partial r$ isn't that much better than a back-of-the-envelope estimate of $\partial u/\partial r$ using characteristic scales. It should be fine to opt for the latter. In fact, the final answer will be the same. •

(103) **2023 E2** As a heads-up, there are two versions of the solution online. To check your work, use the file which includes pictures of the experimental setup. In the other file, the measurements and the graphs are very sloppy, so it's useful only because it also has a marking scheme.

(104) **2024 3B.3** The solution that I've linked in my archives doesn't cover this part correctly. The moments of occultation for A and B in the right panel of the figure differ by $\Delta t = 25\,\text{s}$. We also measure that, upon touching A, the lunar limb needs to shift by an extra 8 mm horizontally to reach B. That distance corresponds to $\omega_\text{M}\Delta t = 14'$. Meanwhile, the two sources are separated by 19 mm, so the angular distance between them should be about $33'$.

(105) **2026 1B.6** A small typo – the second term in the final answer should have an l^2 in the numerator.

IZhO

I've done everything up to 2011 except for some tedious problems that I don't recommend. These are **2017 3**, **2013 3**, **2012 3**. There are also some problems where the graphs are missing, so these can't be worked through: **2017 1.3**, **2015 1**, **2012 1.2**.

(106) **2010 3** There's a nasty typo in the data: the mass of the Δ baryon is $m_\Delta = 1232\,\text{MeV/c}^2$. Also, note that the solution is available only in Russian. •

(107) **2010 3.3a** You're asked to find the energy of the proton, but the solution gives you the energy of the photon instead. For the proton, you should have

$$E_p = \left(\frac{m_p^2 + m_\Delta^2}{2m_\Delta}\right) c^2 = 973\,\text{MeV}.$$

(108) **2010 3.4** The momentum of the π meson in the zero-momentum frame is $q_\pi = 227\,\text{MeV}$ (there's no additional 10^6 factor).

(109) **2010 3.4** The solution works interchangeably with $E^\text{max} = 7 \times 10^{20}\,\text{eV}$ and $E_p^\text{max} = 10^{21}\,\text{eV}$. Don't do that, use only the former value! In that case, the momentum of the π meson is $p_{\pi 1} = 3 \times 10^{20}\,\text{eV/c}$ if it comes out parallel to the initial proton, and $p_{\pi 2} = 2 \times 10^{19}\,\text{eV/c}$ if it comes out antiparallel. The changes in the magnitude of the proton's momentum are then $\Delta p_{p1} = -p_{\pi 1}$ and $\Delta p_{p_2} = +p_{\pi 2}$. ?

(110) **2010 3.6** The solution is wrong, there is no lower bound on the momentum of the photon, given that high-energy protons can allow for any reaction (the optimal case is $E_p \to \infty$, not $E_p = m_p$ as they say).

(111) **2011 1.1** When solving this, you are supposed to take it for granted that it's easiest for the cube to jump up when the puck is at its highest point. However, proving this seems quite difficult. •

(112) **2011 1.1** There shouldn't be a w^2 term when applying energy conservation, only the $(u-w)^2$ is relevant. The equations following this are fine.

(113) **2012 2.2** I don't recommend this problem since its assumptions are shaky. To clarify, the problem statement implies that the change in the internal energy of the gas under the piston is one half of the sum of the gravitational potential energy decrease and the work done by the atmosphere. •

(114) **2012 2.9** I believe that the solution is wrong here. Either the molecular flux statement or the energy conservation statement needs to have the right hand side reversed. This is because the flux equation assumes that the piston is descending with speed u, while the energy equation assumes it's ascending with speed u. Following the correction, I get $T_3 = 44\,\mathrm{K}$ (suspicious). ?

(115) **2013 1.3.1** All the calculations are wrong after a certain point. The correct values are $\beta = 30.7°$, $\alpha - \beta = 24.1°$, $|AO| = 0.82\,\mathrm{mm}$, $|OF| = 0.37\,\mathrm{mm}$, $|DF| = 0.79\,\mathrm{mm}$, $|EO| = 0.58\,\mathrm{mm}$. Their drawing is qualitatively correct, but the dimensions will need changing.

(116) **2013 1.3.2** In spite of the above, their value for the exit angle γ is still correct. However, the small triangles projected on the screen should point outwards rather than inwards.

(117) **2014 1.1** The numerical value for the efficiency is $\eta = 0.53\,\%$.

(118) **2014 1.2** There's a typo in the final answer for Q in the second solution. The answer in the first solution is fine.

(119) **2014 2.6** Note that the maximum height includes the ascent following the exhaustion of the fuel. I don't think that the answer is obvious like they claim in the solution. It's better to get a general answer for H in terms of μ, and then analyse.

(120) **2014 2.9** The answer is $m_0 = 10^{14\,317}\,\mathrm{kg}$.

(121) **2016 2.3.3** The equation should be written down only in terms of h_0, α, and R.

(122) **2016 2.3.4** To make the problem statement more concrete, assume that the height of the tube is almost equal to $R/\tan\alpha$. This will guarantee the existence of equilibrium configurations (for appropriate values of α). •

(123) **2016 3.7** Just as a warning, the solution here involves drawing a tangent, which can possibly lead to large deviations in your numerical values.

(124) **2017 1.2** Their equations are correct, but the final answer for the charge should be

$$q = 32\pi\sqrt{\frac{6}{11}\sigma\epsilon_0 R_1^3}.$$

(125) **2018 3.2.4** The numerical answer in the marking scheme is wrong. The one in the solution is correct.

(126) **2018 3.3.2** Using the same approach (switching to the rest frame of the flow, using Snell's law there, and then switching back), I find

$$\sin\beta = \frac{\sin\alpha}{n} + \frac{v}{c}\left(\frac{n^2-1}{n}\right),$$

implying a constant offset $B_1 = \frac{n^2-1}{cn}$. I got the same result using the formulae on this site. Then **3.3.3** would be wrong as well. The final two tasks are fine. ?

(127) 2020 1.1 The main idea here is to compare the tension forces in equilibrium at midday and midnight. It is precisely these forces that determine the period of the pendulum. The original solution is correct, but it's also incomprehensible. Noninertial frames are tricky, and they haven't talked about the Coriolis force at all. I'd advise you to instead work in an inertial frame and follow the approach in IPhO 1992 1. However, in this problem you need to write down the expression for the tidal force to second order, because the first order terms will cancel out, implying $\varepsilon = 0$.

(128) 2020 1.3.1 The thickness of the border of the triangle should be $\Delta r = 2r_1 = 2\,\text{mm}$.

(129) 2020 1.3.2 The thickness of the border of the star should be $\Delta r = 3r_2 = 0.3\,\text{mm}$.

(130) 2021 1.1 The formulae for the frequencies are correct, but the values should be $\omega_1 = 6.14\,\text{rad/s}$ and $\omega_2 = 10.20\,\text{rad/s}$.

(131) 2022 1.3 The final answer should be

$$Q = \frac{8\sqrt{2}\varepsilon_0 mg}{\sigma}.$$

In going from Equation 5 to Equation 6, they've missed an extra factor of $\cos\beta$. This means you will need to find $\int_{-\pi/2}^{\pi/2}\cos^2\beta\,\mathrm{d}\beta$ rather than $\int_{-\pi/2}^{\pi/2}\cos\beta\,\mathrm{d}\beta$.

(132) 2022 2.9 The answer for α in the marking scheme is wrong. The solution has the right one, $\alpha = 0.014\,\text{K}^{-1}$.

(133) 2022 3.6 The calculation is off by a hundred; The time is $\tau = 1.27\times10^{10}\,\text{s}$.

(134) 2023 3.9 There is a typo in the formula for T. We should have $T \ll \frac{eV_0}{k_B} = 8.70\times10^5\,\text{K}$.

(135) 2024 2 The radius of the Earth's orbit is missing its units, $R_0 = 1.5\times10^8\,\text{km}$.

(136) 2025 1.1 The problem statement is too vague. You're supposed to work in an approximation where the Moon is on a circular orbit around a static Earth. Treat the approach to the Moon in two stages. Initially you're only under the gravitational influence of the Earth. And when you're about to cross the lunar orbit, you're only influenced by the Moon.

(137) 2025 1.1 The solution is wrong because it doesn't account for conservation of angular momentum, which makes it impossible to approach the Moon exactly head-on. Instead, the maximum velocity will be obtained when we launch the spacecraft such that it grazes the Earth. The tangential velocity of the spacecraft near the lunar orbit is $v_\tau = v_{2E}(R_E/r)$, while the normal velocity is $(u^2 - v_\tau^2)^{1/2}$. Hence, it approaches the Moon with a velocity of

$$V_{\max} = \sqrt{(v_M + v_\tau)^2 + (u^2 - v_\tau^2)} = \sqrt{3v_M^2 + 2v_M v_\tau}.$$

This leads to $v_{\max} = (V_{\max}^2 + v_{2M}^2)^{1/2} = 3.05\,\text{km/s}$. The minimum velocity can be obtained the same way, but this time the tangential velocity of the spacecraft is collinear with that of the Moon:

$$V_{\min} = \sqrt{(v_M - v_\tau)^2 + (u^2 - v_\tau^2)} = \sqrt{3v_M^2 - 2v_M v_\tau}.$$

The final answer is $v_{\min} = (V_{\min}^2 + v_{2M}^2)^{1/2} = 2.93\,\text{km/s}$.

(138) 2025 2.7 Provided that the heat loss mechanism is related to heat condution, I believe that the expression for the power should be $P \propto R^1$. This is because $P \propto \kappa S\left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)$, where S scales as R^2, but also $\mathrm{d}r$ scales as R. In that case, $\tau = \tau_0\left(\frac{R_E}{r_0}\right)^2 = 1.25\times10^{12}\,\text{yr}$.

(139) 2025 2.10 There's a factor of $\frac{3}{4}$ missing from the final answer, and the corrected numerical value is $\Delta T = 2.36 \times 10^{-2}\,\mathrm{K}$.

(140) 2025 3.8 One subtle point that the solution glosses over is what happens when $z = 0$. There, the flux from the monopole changes sign, but the current in the loop stays the same nonetheless. This is because of a modification in Ampère's law in the presence of magnetic charges. See this note for more details.

(141) 2025 3.9 This is a bit unclear. They're asking you to find the current when $z = -\infty$.

(142) 2025 3.16 There's an $\hbar^2$ omitted from the final answer.

(143) 2026 1.1 You also need to assume that $\delta L \ll R^2$, otherwise the radius of the roll would change significantly as it unwinds.

(144) 2026 2.11 Assume that η remains the same as before. That is, η is not readjusted so that the pressure is again $p = P_A$ at $V = V_0$.

(145) 2026 2.11 The phrase "when calculating the work performed by the steam, neglect atmospheric pressure" can be interpreted in multiple ways. What they actually want you to do is to ignore the work in the 3–4 branch of the cycle only. In particular, do not set the minimum pressure of the cycle to zero for the 2–3 branch.

(146) 2026 2.11 And after all that, their numerical answer is still wrong. You should get $A_0 = 1645\,\mathrm{J}$. This error propagates to **13**, where $M_{0\max} = 262\,\mathrm{N\,m}$.

(147) 2026 3.4 Pay no mind to the marking scheme, you should get $g_L = 1$ as in the detailed solutions.

(148) 2026 3.8 Planck's law has been written down incorrectly. This renders the problem unsolvable unless you use the right form:

$$\rho = \frac{\hbar\omega^3}{\pi c^3} \frac{1}{\exp\left(\frac{\hbar\omega}{k_B T}\right) - 1}.$$

(149) 2026 3.10 Calculation error, $P = 1.05\,\mu\mathrm{W}$.

(150) 2026 3.11 Likewise, $P = 1.02\,\mathrm{nW}$.

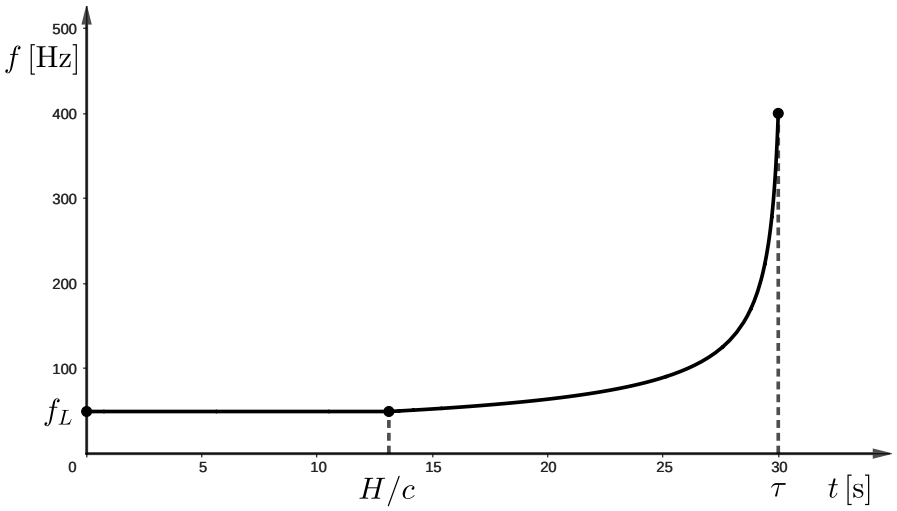
USAPhO

(151) 1997 B1 In part **E**, the assumption that there's no induced field on the axis of symmetry is unphysical. The electrons rotating at any given radius are like the current in a solenoid, meaning that a layer of plasma at r_0 produces a homogeneous field for $r < r_0$, whilst its contribution for $r > r_0$ is zero. To obtain $\mathbf{B}(r)$, you should use Ampère's law for a rectangular loop between r and R, and the answer is $\mathbf{B}(r) = -\mu_0 e n_0 \omega \frac{R^2 - r^2}{2} \hat{\mathbf{k}}$. For part **F**, it follows that $\frac{F_{\mathrm{mag}}}{F_{\mathrm{el}}} = \frac{v^2(R) - v^2(r)}{c^2}$.

(152) 1998 B1 The original solution to part **E** doesn't account for the finite travel time of sound. First, note that the height of the helicopter is $H = \frac{g\tau^2}{2} = 4500\,\mathrm{m}$. Then, following the same approach as in 2008 A4, we should find that $f = f_L$ for $t < \frac{H}{c}$, and

$$f = \frac{f_L}{\sqrt{1 - \frac{2g}{c}\left(t - \frac{H}{c}\right)}} \qquad \text{for} \quad \tau \geq t \geq \frac{H}{c}.$$

The graph of the frequency should look like this:



(153) 1998 B2 In part **A**, the number of protons is $Z = 56$. In the original solution they plug in $r_1 = r_{\max}$, but in reality $r_1 = r_{\max}/\beta$. This error propagates to part **B**, where I get $\lambda = 5.8 \times 10^{-9}\,\mathrm{m}$.

(154) 2000 B2 For part **G**, append the line "assuming that the other wheel does not start slipping in the meantime" to the original problem statement. •

(155) 2004 B2 The answer in part **F** needs a sign change: $\Delta \mathbf{L}_{\mathrm{EM}} = +\frac{1}{2} Q \mu_0 n I (b^2 - a^2) \hat{\mathbf{z}}$. Only then does it become apparent that the total angular momentum of the system is conserved.

(156) 2006 B1 In part **B(ii)**, the equation for $y(t)$ should be

$$y(t) = \frac{mgL}{(BX)^2} \left(1 - \cos\left(\frac{BX}{\sqrt{mL}} t\right)\right).$$

The height D doesn't have anything to do with it.

(157) 2006 B1 There are multiple answers for part **B(v)**:

$$t = (2n+1)\frac{\pi\sqrt{mL}}{BX}, \qquad n \geq 0.$$

(158) 2007 A2 There's a minor typo in the final answer for part **A(ii)**. The exponent is 3/5, not 5/3.

(159) 2007 B2 To clarify, in part **A(iii)** you're being asked to find the time-averaged magnetic field. The instantaneous magnetic field has a different form, which can be obtained using the Biot-Savart law for a single charge $-e$ moving with speed $\omega_0 R$. This would correspond to a magnitude $B_e = \frac{\mu_0 e \omega_0 R}{4\pi(R^2+z^2)}$, akin to a monopole rather than a dipole. But note this is inaccurate anyway, because a single electron does not constitute a steady current, and so technically you're not allowed to use Biot-Savart. See the chapter on magnetostatics in Griffiths for more details. •

(160) 2007 B2 Parts **B(iii)** and **B(iv)** are wrong, because the EMF must depend on the time interval Δt when switching on the magnetic field. If this is done at a constant rate, we have $\mathcal{E} = -\frac{B_0}{\Delta t}\pi R^2$, but then the next part of the problem wouldn't make any sense. The original solution follows the unfounded assumption that Δt equals exactly one orbital period of the electron. •

(161) 2025 B1 A typo in the solution of part **B**, the answer is $v' = v/4$.

(162) 2026 B1 In part **C(i)** you need to find only the initial ratio of the pressures. •

(163) 2026 B3 It is crucial to clarify that the solenoid is not in contact with the cylindrical shell. The phrase "tightly wound" only implies that the solenoid also has radius a and length l. •

NBPhO

(164) 2003 4.4 The graph given here represents $U_{\text{in}}(t)$.

(165) 2003 5.4 In the solution, the expressions for the forces should be divided by 2 in both cases.

(166) 2004 5.1 The correct answer is $\varphi = \arctan\left(\frac{b}{a}\right)^2$.

(167) 2004 7.2 To be more precise, find an equation relating p_0 and T_0 to the density ρ and temperature T of the air coming out the top of the pipe. Here you will also need $g = 9.8\,\text{m/s}$.

(168) 2004 7.3 Assume that the heat sink is small in size and lies at the bottom of the pipe. When the air passes through the heat sink, its thermodynamic parameters undergo a much more significant change compared to what happens further up the pipe.

(169) 2004 7.4 Assume that the changes in the kinetic energy and the gravitational potential energy of the air flow are negligible compared to the change in its internal energy.

(170) 2004 7.5 Strictly speaking, the answer to part **2** should be $p_0 - \rho_0 gL = \frac{\rho}{\mu}RT$. You will need to demonstrate that $\frac{\rho-\rho_0}{\rho_0}$ and $\frac{T-T_0}{T_0}$ are much greater in magnitude than $\frac{\rho_0 gL}{p_0}$, but after that you'll converge with the original solution. Except they make a mistake in the algebra right at the end. The answer should be

$$T = T_0\left(1 + \left(\frac{\gamma-1}{\gamma}\frac{P}{p_0 S}\right)^{2/3}(gL)^{-1/3}\right) = 325\,\text{K}.$$

(171) 2004 8.3 There's a small typo in the units of the final answer, $p = 2.08 \times 10^{-6}\,\text{kg m/s}$.

(172) 2005 4.1 A missing ρ: $p_2 = p_0 - \rho gh + \frac{1}{2}\rho\omega^2 r^2$.

(173) 2005 4.3 A factor of 2 has been dropped in the calculations, the right answer is

$$\omega_{\max} = \frac{\sqrt{2}}{r}\sqrt{\left(\frac{S_1}{S_2}\right)^2\left(\frac{p_0 - p_k}{\rho} - gh\right) + gh}.$$

(174) 2005 5.3 There's a calculation error, you should get $\nu' \approx (5000 \pm 50)\,\text{kHz}$.

(175) 2005 6.4 A typo in the final answer:

$$\omega^2 = \frac{1}{2}\left((\omega_0^2 + \omega_1^2) \pm \sqrt{\omega_0^4 + \omega_1^4 - 2\omega_0^2\omega_1^2\left(5 - 4\frac{X_0}{X_1}\right)}\right).$$

For a more detailed solution to this problem, see Kevin Zhou's handouts.

(176) 2006 7.4 Assume that all molecules have a thermal kinetic energy of exactly $k_B T$.

(177) 2007 3.1 The diagram is misleading, you'll need to change the $U(t)$ there to $-U(t)$. You can then interpret $U_B(t)$ and $U_C(t)$ from the problem statement as the voltage changes at B and C, respectively. The essence of the setup here is that the velocity of the electrons changes instantaneously at B and C. The segments BC and CD are covered with constant speed because time-dependent potentials can change without implying any force.

(178) 2007 3.4 Here you're given T, but you don't know explicitly what U_m is. Also assume that the period of the signal is $T > a\sqrt{\frac{m}{2eU_0}}$.

(179) 2007 3 The original solution is very terse and therefore difficult to understand. For part **3**, I also think it's wrong. I've rewritten the whole solution from scratch, see here.

(180) 2008 9.2 There are some errors in the algebra:

$$v_t = v_0\sqrt{\frac{\alpha\beta}{\alpha+\beta}} = 30.2\,\mathrm{km/s}, \qquad v_r = v_0\sqrt{\frac{2(\alpha-1)(1-\beta)}{\alpha+\beta}} = 16.7\,\mathrm{km/s}.$$

(181) 2008 9.4 Somewhat more accurately,

$$w = \sqrt{u_r^2 + u_t^2 + \frac{2gR_0^2}{R_0+h}} = 20.0\,\mathrm{km/s}.$$

(182) 2010 2 While the problem is nice and the solution is correct, I found it very, very difficult to understand. I've given the solution a complete rewrite here, which should hopefully make things easier. I've also fixed a minor typo in the answer to the second subpart.

(183) 2010 9.1 Assume that the currents are initially zero. Also, both the input and the output are connected such that the upper terminals sit at a higher potential relative to the lower terminals. •

(184) 2011 7 Assume that the piston is pulled up very slowly. •

(185) 2013 3.3 In this part and the next, the numerical values in the solution are inaccurate. Swap them as follows: $11.2\,\mathrm{km/s} \to 11.6\,\mathrm{km/s}$, $3.25\,\mathrm{km/s} \to 2.62\,\mathrm{km/s}$, $5.98\,\mathrm{km/s} \to 5.66\,\mathrm{km/s}$, $2.44\,\mathrm{km/s} \to 2.11\,\mathrm{km/s}$.

(186) 2015 2.1 All that really matters is that the transparent rings give you constructive interference, so for an arbitrary offset α any answer along the lines of

$$r_m = \sqrt{((f+\alpha)+m\lambda)^2 - f^2}$$

is technically fine. If you want to follow the original solution, the first transparent ring there has an optical path difference of $\lambda/2$ with the centre. Label this ring with $m = 0$.

(187) 2015 2.3 The total duration of the pulse is

$$\tau = \frac{1}{c}\left(\sqrt{\left(\frac{d}{2}\right)^2 + f^2} - f\right) = 3.9\times10^{-11}\,\mathrm{s}.$$

(188) 2021 3.3 The balloon does not have a fixed radius r; the value in the problem statement corresponds to the equilibrium position. Assume that the membrane of the balloon is a freely deformable non-elastic sack. •

(189) 2021 3.3 Also correct is

$$m = \frac{4\pi r^3(M-M')P(R-H-l)}{3R_GT(R-H)} = 110.7\,\mathrm{kg}.$$

(190) 2022 7.3 There shouldn't be a 4 in the denominator of the final answer.

(191) 2022 8.5 The original solution doesn't account for the axial tilt of the Earth, which is wrong. And they themselves told you that it's possible to see Venus at midnight in the summer, which contradicts the question! To patch up all of this, assume that we're observing from Tallinn on the September equinox. Solving this will require you to set up a spherical triangle. You should get an answer of 4 h 56 min.

(192) 2024 4 Clarification – the totality lasts $t_0 = 2\,\text{min}$ for those observers at $\lambda = 28.5°$ that are best-positioned to observe the eclipse.

(193) 2024 6D In Solution 4, the radial component of the normal force is not $\frac{mv_\perp^2}{r}$. The normal force N balances the normal component of the centrifugal force, so $N = \frac{mv_\perp^2}{r}\cos\theta$. We plug this into $N\sin\theta = m\dot{v}_\parallel\cos\theta$ (from Solution 3) to find $\dot{v}_\parallel = \frac{v_\perp^2}{r}\sin\theta$. You can follow the official solution after that, but note that the final integral doesn't have anything to do with the area of a circle. That would be $\int_0^1 \sqrt{1-u^2}\,\mathrm{d}u$, which evaluates to $\pi/4$, not $\pi/2$.

InPhO

(194) 2014 7E One of the terms in β has an extra factor of $5/3$:

$$\beta = \frac{1 + f^{1/3} + (5/3)f + f^2 + f^{5/3}}{(1+f)^{5/3}(1+f^{1/3})}.$$

GPhO

(195) 2019 2C When solving this, assume that the entropy change due to the ionisation is negligible. Then, all the work done by the external pressure goes towards raising the internal energy of an ideal gas.

(196) 2019 2C.3 The correct answer is $d = R$.

(197) 2019 2D.3 They've missed the factor of $\frac{3}{2}$ in the final answer, you should get $T_m = \frac{p_e}{6p_0}T_0$.

(198) 2019 2D.3 The procedure they use for finding r_m seems dubious to me. The process might be adiabatic, but it's still far from quasistatic, so $TV^{\gamma-1} = \text{const}$ is inappropriate here.

(199) 2022 1B.2 A factor of 2 has been omitted: $v_\text{th} = \sqrt{\frac{3}{2}}v_0 = 280\,\text{km/s}$.

(200) 2022 2.2 You should assume that that the solar panels are evenly distributed across the surface of the Earth. Note that the solutions file compares the total area of the panels with the Persian Gulf rather than the Empty Quarter, but this is a minor inconvenience.

ISPhO

Nothing here yet!

Contributing to the list

There are many errors missing from this file, and a single person can't hunt all of them down. This is where I ask for your help! If you have found an error, please email me so that I can add it to the list. Borrowing Donald Knuth's idea, I will award physics money (i.e. Joules) for your troubles, as follows:

- **Clarifications.** Worth 5 J. If you think that a problem statement is too ambiguous for someone to get the problem right the first time around, I can try to tidy it up here. Be explicit in what it was that had you confused.
- **Verifications.** Worth 10 J. There are some errors here that I am not certain about. I've marked them with a **?**. I'd like someone else to double-check those. Message me with the number of the error (e.g. **(17)**) and attach some working which supports or disproves what's written down in the list. It doesn't have to be neat, just legible.
- **Wrong solutions.** Worth 10 J. Some problems are correctly stated, but there are major issues with their solutions. What I count as an error is something which leads to a wrong final answer, either in the formula or in the numerical value. For example, a minus which disappears in one line of the solution but reappears in the next is fine with me – this sort of typo is quite common and not too harsh on the reader. Should you notice a significant error, please:
  1. Explain why the official solution is wrong.
  2. Show me what the correct answer is.
- **Wrong problems.** Worth 15 J. Occasionally there are problems which are so wrong that one cannot patch up the solution and call it a day. One way this can happen is when a problem author forgets about a key physical effect, and the setup actually does things which are completely different from what the problem statement hints at (e.g. instability instead of oscillations). If you think a problem is wrong, please outline why. There should be enough detail so as to convince a fellow student.

Keep in mind that I am only tracking the competitions listed above, that is, IPhO, EuPhO, APhO, IZhO, USAPhO, NBPhO, InPhO, GPhO, and ISPhO. I'm generally rather slow to reply, but still, you could send me a reminder if I haven't addressed your query within a month.

Energy balance

Additions to the list are tracked and credited. If you want to stay anonymous, that's alright too!

Teo Kai Wen ..... 140 J
▷ (17), (23), (40), (43), (65), (66), (67), (69), (151), (159), (160), (186), (187), (190), (195), (196), (197)

Nikola Kočović ..... 40 J
▷ (143), (165), (166), (170), (171)

Mohammed Alsawafi ..... 40 J
▷ (2), (80), (81), (91), (95)

Kevin Zhou ..... 35 J
▷ (1), (172), (173), (175)

Sirawish Musikasopon ..... 30 J
▷ (51), (52), (103)

Smbat Poghosyan ..... 20 J
▷ (58), (59), (62)

Feodor Yevtushenko ..... 20 J
▷ (76), (78)

Lachlan McGinness ..... 20 J
▷ (96), (98)

Atanas Gochev ..... 20 J
▷ (156), (199)

Pavle Žužić ..... 15 J
▷ (3), (4)

Murad Bashirov ..... 15 J
▷ (112), (182)

5929 ..... 10 J
▷ (155)

Alan Wang ..... 10 J
▷ (158)

Eppu Leinonen ..... 10 J
▷ (193)

David Hayrapetyan ..... 5 J
▷ (26)

Matías Fermín Páez ..... 5 J
▷ (105)

Alex Prodanov ..... 5 J
▷ (127)

. . . and other anonymous contributors.